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RESEARCH ARTICLE



Public responses to hard climate policies in OECD member countries: prevalence of contention at the post-adoption stage

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ABSTRACT

Effective climate action is likely to require hard climate policies (e.g. regulation, pricing, phase-outs), but such policies are often contested. Contention can arise not only during policy development and decision making but also at the post-adoption stage where adopted policies may be challenged, threatening policy durability. However, existing research has not systematically analyzed patterns of actual public responses to hard climate policies following their adoption. This is needed to understand response variety and the prevalence of contention. In this paper, we systematically assess public responses in a sample of 50 national-level hard climate policy packages (covering framework and energy policies) across 21 OECD member countries between 2009 and 2022. We categorize public responses (support, acceptance, non-acceptance, backlash) using a typology involving disapproval and mobilization to assess the prevalence of post-adoption contention among mass publics. We draw on newspaper data specific to each case to detect contention. Findings reveal a relatively low prevalence of post-adoption contentious responses to hard climate policies. Nonetheless, contention is pronounced in some cases and is mainly associated with carbon taxation/pricing although not exclusively. The most contentious responses, labeled as backlash, cluster in the latter half of the study period. Findings suggest a need to analyze drivers of contention among mass publics which may be complex and involve factors spanning policy design, institutional politics, and societal context.

Key policy insights

- We observe a low prevalence of public contention over hard climate policy instruments (e.g. regulation, pricing, phase-outs) at the post-adoption stage within OECD countries between 2009 and 2022.
- Instances of backlash to hard climate policy are relatively rare but can be consequential when they occur. Backlash mostly occurs when policy is seen as too ambitious but can sometimes also occur when policy is seen as not ambitious enough.
- No clear links are found between various attributes of hard climate policies (i.e. instrument type, gradual introduction, country, timing) and the occurrence of public contention.
- Politicians and policymakers should not be axiomatically fearful about the lack of public acceptance of hard climate policies.

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Climate policy; policy acceptance; contentious action; comparative analysis; backlash

1. Introduction

The volume of climate policies introduced in countries across the world has grown substantially over the last 10–15 years (Eskander et al., 2021; Nascimento et al., 2022; Roelfsema et al., 2020; Schmidt & Fleig, 2018).

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Policies range from hard (regulation, prices/taxation, phase-out of technology or fossil fuel production) to soft (information campaigns, voluntary standards) (Schulze, 2021). Hard climate policy is often considered crucial for rapid and large-scale decarbonization but can be politically controversial because it involves distributing economic and social costs (Harrison & Sundstrom, 2010; Jordan & Matt, 2014; Patterson, 2023). Sometimes contentious public responses occur even after adoption, such as when costs become apparent or new challengers arise. Examples of contentious post-adoption situations include rollback of carbon pricing in Australia (2012–2014), the Yellow Vests Protests in France (2018–2019), and rollback of climate policy in Ontario, Canada, along with provincial court challenges against national climate policy (2019–2020). This creates concerns about the risk of backlash or future rollback of climate policy (IPCC, 2022; Patterson, 2023). However, we lack systematic understanding of actual patterns of public responses to hard climate policy after it is adopted.

Addressing this gap requires bringing together insights on policy acceptance and contention at the post-adoption stage of policy. Bourgeoning literature examines ex-ante determinants of public support for climate policy, either in general or for individual policies (Bergquist et al., 2022; Dechezleprêtre et al., 2022; Drews & Van den Bergh, 2015; Kyselá et al., 2019). However, expected patterns of support may not always correspond to actual (ex-post) public responses (Bakaki & Bernauer, 2017; Dechezleprêtre et al., 2022; Douenne & Fabre, 2020; Drews & Van den Bergh, 2015), especially when costs are felt, political debates and struggles shift the meaning of policy, and external events change the impacts and prioritization of policy, for example. Moreover, actual public responses to climate policy remain understudied (Kallbekken, 2023). Scholars both within climate policy (Jordan & Moore, 2020; Stokes, 2020) and beyond (Patashnik, 2008) have observed that struggles over policy often continue after policy adoption. On the other hand, public support might increase following policy adoption as new policy becomes normalized (Drews & Van den Bergh, 2015; Kallbekken, 2023). Hence, it is not necessarily clear which public responses may occur in response to specific hard climate policies, even though we might nonetheless expect the potential for discontent at the post-adoption stage.

We therefore ask the question: How prevalent are contentious public responses among mass publics towards adopted hard climate policies? We systematically analyze post-adoption public responses to hard climate policy in a sample of 50 policy packages across 21 OECD member countries between 2009 and 2022. By 'hard' climate policies,¹ we refer to those that impose costs through 'behavioral constraints or negative economic incentives' on policy recipients (Schulze, 2021, p. 3) backed by the threat of sanction or force (Salamon, 2001). We consider three categories of hard policy instruments: regulation (which imposes limits or constraints on certain forms of behavior), pricing/taxation (which imposes mandatory costs on certain behavior or activities), and phase-outs (which imposes mandatory shifts away from certain forms of technology or production) (Borrás & Edquist, 2013; Pacheco-Vega, 2020; Rinscheid et al., 2022). We categorize post-adoption public responses using a typology spanning disapproval (present/absent) and mobilization (present/absent) to distinguish four possible public responses (backlash, non-acceptance, acceptance, support), drawing on systematically sampled newspaper data (newspapers of quality) in each country. Findings reveal a relatively low prevalence of post-adoption contentious public responses to hard climate policies. When this does occur, it is mainly associated with carbon taxation/pricing, although not exclusively. The most contentious responses (i.e. backlash) cluster in the latter half of the study period, and the role of design features (e.g. gradual phase-in, redistribution) appears ambiguous.

Overall, this contributes to understanding the role of mass publics in shaping policy acceptance (Dermont et al., 2017; Ingold et al., 2019; Kammermann & Dermont, 2018) and durability (Jordan & Moore, 2020) by examining relatively near-term post-adoption public responses (up to 3 to 6 years from policy adoption) complementary to analysis of gradual policy feedback involving mass publics over longer timeframes (Jacobs & Weaver, 2015; Mettler & Soss, 1999; Sewerin et al., 2020; Stadelmann-Steffen & Eder, 2021). We provide evidence on the prevalence of different types of public responses to hard climate policies using a medium-*N* set of cases (i.e. $n = 50$, which reflects a middle ground between a small-*n* [e.g. $n < 5$] or large-*n* [e.g. $n > 100$] sample size). The findings are relevant for understanding both the supply side of climate policy regarding patterns of hard policy employed to date and the demand side in terms of actual public responses, which can help to inform expectations for future policy action.

This article proceeds as follows: Section 2 explains how we delineate public responses and proposes a typology for studying them; Section 3 explains the methodology including case sampling, data collection,

operationalization and analysis; [Section 4](#) presents the results; [Section 5](#) reflects on the findings within ongoing climate policy debates, and [Section 6](#) concludes with policy implications and identifies future research needs.

2. Public responses

2.1. Delineating public responses

We focus on public responses towards specific adopted climate policies, which encompass both policy recipients and wider audiences within mass publics. Public responses are an important aspect of the overall issue of policy acceptance which goes beyond only ex-ante public support (Batel et al., 2013; Wüstenhagen et al., 2007) and involves multiple actors (e.g. mass publics, political representatives, interest groups) and stages of a policy process (e.g. policy development, decision-making, adoption, implementation) (Dermont et al., 2017; Upham et al., 2015). Our focus on public responses reflects a concern with what Wüstenhagen et al. (2007) term the socio-political dimensions of acceptance; that is, whether and how a new policy (which may involve new targets, rules, arrangements or frameworks) is accommodated within a social and political unit. We build on Dermont et al. (2017) in considering the post-adoption stage of policy which overlaps with, but is also distinct from, the implementation stage because post-adoption centres on public responses when policy becomes (or is slated to become) operational rather than on debates over how to bring a policy into operation per se. It aligns with a view of policy acceptance as a process (Batel et al., 2013; Upham et al., 2015; Wüstenhagen et al., 2007) because the post-adoption stage is part of the ongoing process by which acceptance extends over time beyond adoption.

The scope of policy acceptance analysis can include a wide spectrum of public responses ranging from highly negative (e.g. resistance, opposition) to somewhat negative (e.g. disagreement, concern) to ambivalent (e.g. indifferent, uncertain) to somewhat positive (e.g. acceptance, tolerance) to highly positive (e.g. support, affirmation) (following Batel, 2018; Batel et al., 2013; Dermont et al., 2017; Wolsink, 2018). Our primary focus is on negative contentious public responses (both highly and somewhat negative responses which can threaten or undermine adopted climate policy), while also considering this wider spectrum. Contentious public responses may occur relatively immediately after a policy is adopted, such as outbursts of anger or criticism, potentially (but not necessarily) primed by prior debates, or it may occur later when a policy ramps up. Contention might also be triggered by organized interests such as industry seeking to foster discontent in the public arena (Mildenberger, 2020) through media or other campaigns. Contention can include negative sentiment among mass publics conveying disapproval, as well as behaviors demonstrating discontent such as protests, demonstrations, petitions or other public actions.

2.2. Typology of public responses

We develop a typology to characterize mass public responses to hard climate policies. A focus on mass publics acknowledges their central role in policy acceptance and durability through both sentiment (i.e. the presence or absence of disapproval for a policy) and behavior (i.e. the presence or absence of mobilization in regard to a policy). Literature on public support for climate policy makes clear the need to consider sentiment towards a policy not only before but also after policy adoption (Bakaki & Bernauer, 2017; Dechezleprêtre et al., 2022; Drews & Van den Bergh, 2015; Kallbekken, 2023), as the presence of widespread disapproval can strengthen opponents or even undermine support from proponents due to public criticism (Crowley, 2021; Mildenberger et al., 2022; Patterson, 2023). Literature on social movements makes clear the need to consider contentious behavior involving mass publics as key claimants, alongside institutional actors (political parties, interest groups) (McAdam et al., 2001). This can include both organized and spontaneous activities (such as protests, strikes, campaigns or other expressions of discontent) in the public sphere with ‘manifestly [public] political ramifications’ rather than ‘claim making that occurs entirely within well-bounded organizations’ (McAdam et al., 2001, p. 5). Scholars in climate and energy policy increasingly observe contentious behavior as important regarding climate policy (Driscoll, 2023; Gulliver et al., 2021; Jerolmack & Walker, 2018; Sovacool et al., 2022; Temper et al., 2020).

Considering both sentiment and behavior can help to distinguish different types of public responses. This not only reflects the potential for a wide spectrum of public responses as argued by policy acceptance scholars but also helps to structure empirical analysis of these responses by categorizing them according to underlying dimensions; that is, the presence/absence of disapproval (sentiment) and the presence/absence of mobilization (behavior). Moreover, this also helps to connect policy processes (specifically, at the post-adoption stage) with contentious action which is a persistent gap observed by both policy-oriented and social movement scholars (Hacker & Pierson, 2014; McAdam & Tarrow, 2010; Tarrow, 2013). Doing so is important given growing experiences of mobilization for and against climate policy. Mobilization demonstrates a stronger signal than disapproval alone and could thereby have distinct consequences for policy acceptance and durability.

On this basis, we posit a typology to empirically categorize public responses in Figure 1. Contentious public responses involve substantial disapproval over a policy among mass publics but vary as to whether this extends to mobilization or not. A response of ‘non-acceptance’ involves the presence of disapproval but with an absence of mobilization, suggesting some contention but not severe enough to motivate behavioral responses against the policy. This involves criticism and/or opposition (and other forms of conflict inherent to the policy process; Ingold, 2011) which could dissipate over time and transition into acceptance or conversely could lead to policy undermining later. A response of ‘backlash’ involves the presence of disapproval combined with the presence of mobilization against the policy, suggesting substantial contention affirmed through behavioral dissent. This could involve protests and demonstrations as well as other forms of non-compliance or resistance by a substantial number of policy recipients and/or possibly further challenges against the policy (Patterson, 2023; Stokes, 2016). On the other hand, a response of ‘acceptance’ involves the absence of disapproval also with an absence of mobilization, suggesting little or no contention. This indicates that a policy is accommodated through either active compliance (Dermont et al., 2017), or passive acceptance such as when policy recipients are not concerned about, or perhaps even aware of, a policy. While scholars have convincingly argued that acceptance should in principle be intentional rather than merely by default (Batel et al., 2013; Dermont et al., 2017; Upham et al., 2015), we nonetheless make this simplifying assumption here because our core focus is on whether policy is challenged following its adoption (in other words, our anchor is the presence/absence of disapproval). If policy is not challenged, then the net result is acceptance of the policy, although in our empirical approach (Section 3) we specifically allow for the possibility that contention is delayed and thus do account for situations where initial passive acceptance is superseded by contention. On the other hand, we also distinguish affirmative responses through the final category of support. A response of ‘support’ involves the absence of disapproval (which could also be evident directly through signs of approval) combined with the presence of mobilization oriented positively towards the policy. For example, this could include demonstrations or other activities taken by mass publics in support of a policy (Gulliver et al., 2021). While this category necessitates some further interpretation in empirical analysis given the logic of the typology, it is nonetheless distinct from acceptance as it asserts a stronger positive stance towards the policy, and from backlash as it affirms an adopted policy rather than challenges it.

This approach has several benefits. First, public responses can vary beyond binaries of acceptance/non-acceptance or support/opposition (Batel, 2018; Batel et al., 2013; Dermont et al., 2017; Sælen & Aasen, 2023; Scherhauser et al., 2017; Wolsink, 2018). Considering the role of mobilization along with disapproval helps to combine insights from literature on policy acceptance and contentious action, providing a structure to organize different public responses. For contentious (negative) responses, distinguishing between non-acceptance and

		Mobilization	
		Absent	Present
Disapproval	Absent	Acceptance	Support
	Present	Non-acceptance	Backlash

Figure 1. Typology of public responses to climate policy.

backlash helps to avoid labeling all negative public responses the same (e.g. counter to the apparent growing tendency in climate policy debates to readily label criticism as backlash, which arguably should be assessed more specifically). For positive responses, it helps to distinguish between acceptance and support, following calls to do so (Batel et al., 2013; Dermont et al., 2017; Dreyer & Walker, 2013). Furthermore, while the approach anchors on disapproval (in line with our research question), giving less explicit attention to positive responses, doing so allows for the possibility of asymmetric explanations for different public responses. In other words, the factors or conditions explaining a category on one end of an analytical spectrum might not be the inverse of the factors explaining the opposite category. Therefore, while we do not seek to explain the observed public responses in detail, the approach provides a basis for explanatory analysis of contentious public responses as a future step.

3. Method

Using the typology in Section 2, we systematically analyze post-adoption public responses to hard climate policy in a sample of 50 policy packages across 21 OECD member countries between 2009 and 2022. These are democratic, advanced industrial economies which arguably have a global responsibility and resources to advance ambitious climate mitigation policy, and can therefore be expected to have taken steps in this direction over the last decade or so. As democracies we expect these countries to afford opportunity (albeit potentially varied) for public criticism and that contention would therefore be both possible and detectable. The start date is the year of the Copenhagen Climate Summit (COP15) when it could be expected that national climate policy was gaining increasing attention and activity (even in the run-up to COP15) and the end date is the end of the calendar year prior to analysis. This approach helps to advance cross-country comparative analysis of public responses complementary to small-*N* analysis (e.g. single or a few cases) which is more common (Dreyer & Walker, 2013; Fairbrother, 2022, p. 2; Ziegler, 2017), and thereby also helps to provide a balanced view since some countries are highly studied (e.g. United States, Australia) which risks skewing empirical debates.

3.1. Case sampling

We systematically sample hard climate policies from a defined population drawing on the Climate Policy Database (CPD) (CPD, 2023), a global open access repository of climate policies (D'Orazio, 2022; Iacobață et al., 2021; Nascimento et al., 2022; Roelfsema et al., 2020). This database tracks national climate policies in 42 countries and the European Union most closely but also in many other countries and draws on other prominent datasets

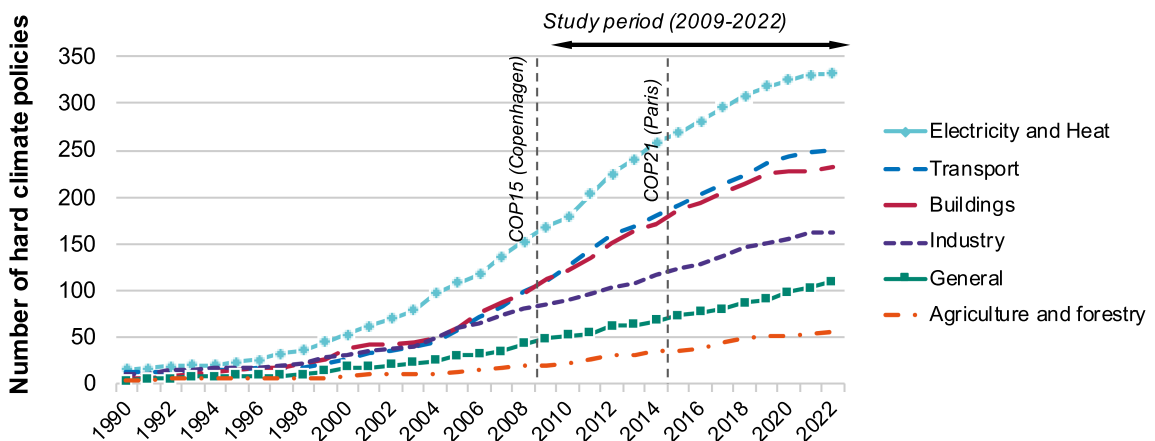


Figure 2. Cumulative adoption of hard climate policies across sectors in OECD member countries over time (accounting for policies ended or superseded).

and reports (Appendix 1). This makes it well suited to systematic analysis of national climate policies in our study. We used the latest complete version of the CPD at the time of analysis (February 2022) and manually added policies from the rest of the 2022 year (Figure 2). This sample frame allows us to avoid sampling on the dependent variable (i.e. to avoid selecting cases based on the outcome – the presence/absence of contention) (following McAdam & Boudet, 2012) enabling the assessment of the prevalence of different types of public responses, especially contention. We treat public responses empirically in reference to a policy itself, not whether the response reflects preferences for or against climate change mitigation per se, to avoid making assumptions about policy appropriateness.²

Given our focus on hard climate policy, we first identify hard policy instrument types within the CPD by screening and re-categorizing instrument identifiers used in the database (see Appendix 1) to align with our definition of hard policy (Section 1). This includes policies that regulate or limit certain activities, impose taxes or charges, or phase-out certain activities. This is important because policy instruments are often examined individually and with different levels of attention, and some instruments such as carbon taxation/pricing receive more attention than others (Fairbrother, 2022; Maestre-Andrés et al., 2019; Umit & Schaffer, 2020). Hence, our approach contributes to examining comparative patterns across different instruments within a certain policy type (i.e. hard policy) (Bergquist et al., 2022; Schulze, 2021).

We then apply a sequential case selection process (see Appendix 2). We first select two sectors (as per CPD terminology): (i) general frameworks (policies that apply across sectors and/or frame or underpin sectoral policies) and (ii) energy and heat (policies related to energy supply and infrastructure) (Figure 2). We select general frameworks due to its broad coverage, and energy and heat as the largest domain of activity, thus both are highly relevant to emissions reduction and decarbonization (Ge et al., 2020; Lamb et al., 2021; OECD, 2022b). We then screen to check that hard instrument types are central (rather than peripheral) elements.³

We then consider whether the policies impose costs on mass publics. Although our focus is on mass publics, interest groups can impact debates in the public sphere and thus public responses to policy. Hence, we consider policies with target audiences that are either mass publics or industries if a policy could readily lead to companies transferring costs to customers (e.g. energy supply) leading to contention in the public sphere. This approach echoes Soule and Earl (2005) who consider private organizations fostering public contention. We then consolidate individual policies into policy packages where relevant when there are several closely interrelated policies with common goals that may not be readily distinguished by mass publics (e.g. Australia's Clean Energy Act (2011) and its spin-off policies: Carbon Pricing Mechanism (2012), Clean Energy Future Plan (2012), Carbon Pollution Reduction Scheme (CPRS) – ETS (2012)), to be empirically meaningful when studying public responses. This leads to the final sample of 50 cases across 21 countries (see Appendix 3).

3.2. Data collection

The object of analysis within the cases is public responses to adopted hard domestic climate policies. We consider public responses with resonance at the national level. We assume that valid evidence of public responses, especially contention, will be detectable in newspapers of quality⁴ because contention is likely to readily attract coverage (Robertson et al., 2023; Vliegthart et al., 2011). We also scrutinize public responses that are detected for relevance to national political debates over the climate policy in question. This approach is chosen for several reasons: first, general audience mass media such as newspapers provides a 'master arena' for capturing public discourse (Gamson, 2004, p. 243); second, this approach supports accuracy by drawing on trustworthy newspapers of quality; third, it supports comparability across cases through using a broadly comparable data type.

We identify newspaper sources on a country-by-country basis to select quality newspapers with national scope (see Appendix 4). We first derive a list of four newspapers per country drawing on the compilation of the European Social Survey supplemented by our analysis of a country's newspaper landscape, previous studies using a similar approach (e.g. Peterson, 2021) and consultation with scholars knowledgeable about each country. We then apply a systematic search procedure (see Appendix 5). We first iteratively derive a search string involving theoretical considerations (considering the operationalized typology), empirical considerations (e.g. plain-language terms, linguistic, and contextual suitability) and application considerations (e.g. specificity and noise in search results) (see Appendix 5.1), in accordance with typical procedures for

systematic qualitative searches. The final search string is: ('Policy name' OR [colloquial variation of policy name]) AND (protest* OR demonstrat* OR mobili* OR contest*). We intentionally keep the search string focused on detecting general forms of mobilization to enable later distinction between mobilization against or for the policy.

We then conduct the search using the database NexisUni on a case-by-case basis (see Appendix 5.2). We begin by using two newspapers per country and then extend this to four if sufficient data is not identified in the first round. The timeframe for each search is anchored by the date of policy decision (as recorded in the CPD), with the search period defined as one year prior to the policy decision and three years afterwards. In case of a time lag between the policy decision and the date of implementation, and/or gradual implementation, the search period is extended by three years (i.e. up to 6 years in total from the policy decision). This is a relatively generous approach to ensure that we do not miss relevant public responses arising later in time, such as where policy is challenged in a subsequent election.

The number of newspaper articles collected is highly variable across the cases. For nearly half of policy cases, the number of articles ranges between 1 and 20, while for 15 policies there is no data found in the NexisUni database. To further scrutinize these apparent gaps, and since there is unequal coverage of newspapers across different languages in the database, we also conducted an additional Google search using the same keywords and date range (yet open-ended concerning sources; scanning the first 3–4 pages) to try to detect any further evidence of contention not detected in NexisUni. This did not reveal additional evidence of contention. As a further check, we applied the same search using the Factiva database (Blatchford, 2020; Buntain et al., 2023), which confirmed our original search results (see Appendix 6).

3.3. Operationalization and analysis

We operationalize the typology in Figure 1 using measures to assess the presence/absence of each variable (i.e. disapproval, mobilization). We derive this approach by considering how these variables could be empirically detectable in the context of climate policy, also drawing inspiration from social movement literature concerning ways of measuring public contention (e.g. level of abstraction, use of composite measures, scoring approach) (e.g. McAdam & Boudet, 2012) and use of newspaper data to do so⁵ (Earl et al., 2004; Soule & Earl, 2005). We use an ordinal scale for each measure (0–3) and combine the measures by taking the average of their scores to assess intensity of disapproval and mobilization, respectively. We then apply a (conservative) threshold to identify the presence (i.e. ≥ 2) or absence (i.e. < 2) of disapproval and mobilization, respectively (see Appendix 7). We do this because even if disapproval and/or mobilization is present to a low extent in a certain case, our focus is on detecting contention, in which we want to have confidence.

To detect the presence of disapproval, we use two measures: (i) policy dis-favourability (i.e. dis-favourability of the specific policy, where: 0 = no evidence of disapproval, 1 = limited and/or sporadic dis-favourability, 2 = notable dis-favourability which might vary over time, 3 = widespread and sustained dis-favourability) and (ii) government dissatisfaction (i.e. general dissatisfaction with incumbent government beyond the specific policy, where: 0 = no evidence of dissatisfaction, 1 = limited and/or sporadic dissatisfaction, 2 = notable dissatisfaction which might vary over time, 3 = widespread and sustained dissatisfaction). To detect the presence of mobilization, we use three measures: (i) number of participants (i.e. multiple participants involved in mobilization activities (Earl et al., 2004; Soule & Earl, 2005), where 0 = absent, 1 = 100–999 participants, 2 = 1000–9999 participants, 3 = 10,000 + participants), (ii) directed action (i.e. salient forms of targeted mobilization for political attention), which is a compound measure involving: (a) frequency (i.e. sustained mobilization activities over time, where 0 = absent, 1 = one activity, 2 = two consecutive activities, 3 = three or more consecutive activities), (b) variety (i.e. different types of mobilization activity such as protest, demonstration, or petition, among others, where 0 = absent, 1 = one type, 2 = two types, 3 = three or more types) and (c) attention (i.e. whether mobilization receives attention among policymakers, where 0 = no apparent policymaker attention, 1 = some policymaker attention, 2 = moderate policymaker attention, 3 = high policymaker attention) and (iii) goal specificity (i.e. relevance of the claim towards the specific policy, where: 0 = non-relevant claim, 1 = general claim about climate change, 2 = general claim about climate policy, 3 = claim specifically about the policy). Appendix 7 provides further information on the scoring and coding procedure and two illustrative examples of how this was conducted.

We interpretively code the collected newspaper data by applying the measures using Nvivo (qualitative data analysis software). We first screen the newspaper articles for relevance to retain only those actually pertaining to the case (summarized in Appendix 8). For cases with fewer than 100 newspaper articles, we code all of them (excluding those that are irrelevant after individual screening). For cases with 100+ newspaper articles, we individually screen the articles (based on title and preview) selecting up to 150 most relevant ones. Coding was performed by one coder over a relatively short period of time (12 weeks) following data collection. A pilot coding phase comprising approximately 10% of the data gathered was performed to deliberate on the coding strategy among both authors and make necessary adjustments. The full coding was then performed by the same single coder (first author) in consultation with the second author. Coding involved applying the measures as qualitative codes (labels) to identify relevant data components (e.g. sentences, reported data such as polls and events, statements concerning public responses) to build up a base of evidence to score each measure. Furthermore, if a certain case also showed public responses in two directions (i.e. for and against a policy) which occurred in a small number of cases, then we also coded and scored both responses for a fuller interpretive picture, although this does not detract from the assessment of the case because our focus is on positively detecting the presence of contention. Throughout the coding process, there was a constant reflection back to the previously coded cases to ensure consistency. This follows a typical deductive qualitative coding approach. Finally, we compile the scores for the presence/absence of disapproval and mobilization in each case (rounded to the nearest whole number), and thereby categorize each cases against the typology of public responses (see Appendix 8).

An important limitation is that contention might not always be reported equally in newspapers. We sought to mitigate this issue in several ways including searching in native languages using linguistically and contextually appropriate search terms as far as we could ascertain, focusing on a relatively politically diverse pool of newspapers of quality, and for cases with few results expanding the number of newspapers and conducting additional open Google searches to try to detect any missed contention. Nonetheless it is possible that we still missed some important instances of contention in the cases studied especially due to newspaper nonreporting or due to contention at other levels (e.g. sub-national, local). Another limitation is that due to our focus on positively detecting contention, we assume that cases lacking the evidence of disapproval are categorized as Acceptance even if passive. However, some instances of non-contention might involve policy indifference/unawareness with latent potential for contention.

4. Results

Results reveal a relatively low prevalence of contentious public responses to adopted hard climate policies (12%) while most policies were accepted by mass publics (78%) and occasionally (actively) supported (6%) (Figure 3). Clear patterns between the type of public response and policy instrument type are somewhat difficult to identify. Although, it is noteworthy that a substantial proportion of all three hard policy types locate under acceptance. This is perhaps counterintuitive especially for pricing/taxation policies which are often seen as unpopular and thus with ongoing risk of contention. Yet, on the other hand, there were no pricing/taxation policies actively supported by mass publics while this instrument was prominent among backlash cases.

Policy instruments related to carbon taxation/pricing are the most widely used (56%), followed by regulation (34%), while phase-outs represent only 10%. This distribution suggests that financial and regulatory intervention have been the focus of hard policies to date. Phase-out policies, understood as policy interventions ‘that deliberately seek to terminate one or several socio-technical elements in a gradual or stepwise process’ (Rinscheid et al., 2022, p. 225) are used more rarely, but might increase in coming years.

Temporal patterns of public responses to hard climate policies are somewhat ambiguous (Figure 4). A possible pattern is the clustering of backlash mostly in the latter half of the period studied (i.e. one case pre-2015; five cases post-2015). This might suggest that contentious public responses have increased in the aftermath of Paris Agreement due to anticipated increases in climate policy ambition (Raiser et al., 2020). But the apparent uptick in contention could also be linked to a wider atmosphere of growing populism and/or concerns about economic inequality in various countries within which climate policy becomes swept up (Lockwood, 2018). In

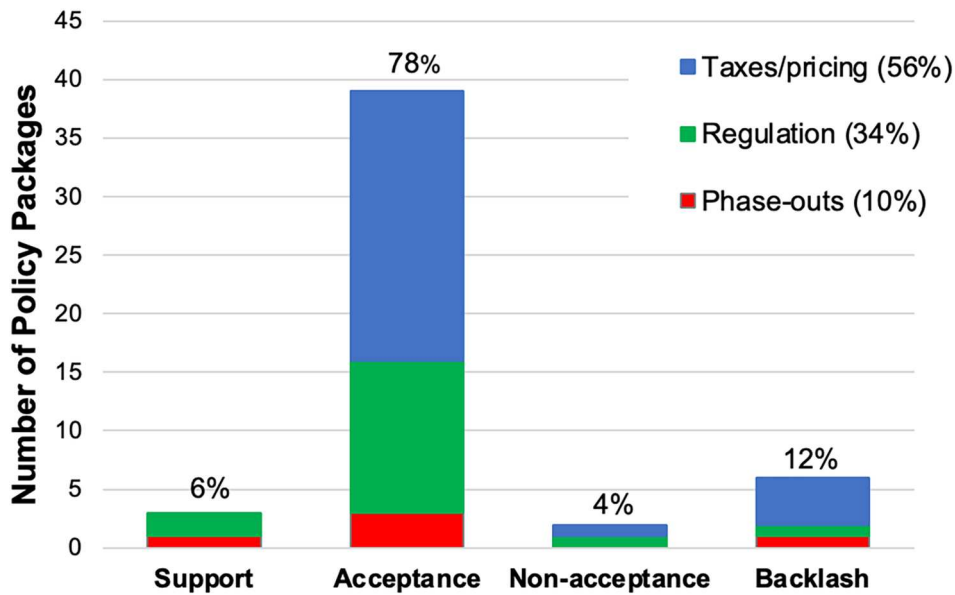


Figure 3. Public responses for different policy instrument types in the sample.

most cases, public responses coincide with the time of policy adoption. However, some cases experienced mobilization up to six years after the policy was introduced (i.e. three cases of backlash; one case of support). This could be due to gradually increasing costs that trigger response at a certain moment (backlash) or eroding policy stability that triggers proactive policy support.

4.1. Support

Three policies encountered public support: the Energy Transition Law (2015) in Mexico, the Clean Power Plan (2014) in the United States, and the prolongation of prohibition of nuclear energy production (2011) in Italy.⁶ These cases involve support in varying situations. Mexico's Energy Transition Law (2015), a regulatory policy, had been endorsed in 2014 by the Chamber of Deputies. But when its passage began to appear stalled,

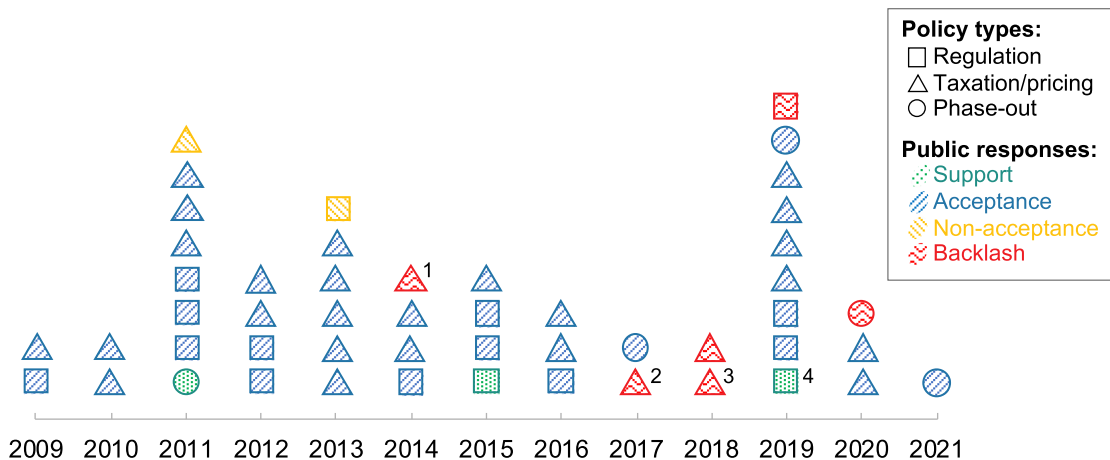


Figure 4. Temporal distribution of public responses to hard climate policies.

nationwide protests arose to affirm support for energy transition involving multiple environment, development, and indigenous organizations, demonstrating a linking of concerns over climate policy and socio-economic development/livelihoods. The US Clean Power Plan (2014), a regulatory policy, was targeted for repeal by the Trump administration in 2017 (accomplished in 2019) to be replaced by the Affordable Clean Energy (ACE) Rule which would relax environment protection standards and allowing states more leniency on emission reduction targets. Public support arose to try to defend the Clean Power Plan through legal challenges and political party positions. The prolongation of prohibition of nuclear energy production (2011) in Italy, an affirmation of an existing phase-out decision, was questioned when the government proposed to increase the use of nuclear power for electricity to mitigate increasing energy costs (Bersano et al., 2020; Esposto, 2008). However, this proposal was strongly rejected by voters in a referendum in June 2011, affirming support for the previous phase-out decision.

These cases show different reasons for mass public support for climate policy. For Mexico's Energy Transition Law (2015), support arose when the policy process stalled reflecting a response to concerns about delays and obstruction. For the US Clean Power Plan (2014), support arose when a new government acted to repeal the policy, representing an affirmative response to protect a prior policy win. For the nuclear energy phase-out in Italy (2011), support for a continued prohibition of nuclear energy production was motivated by fear of future disasters suggesting mobilization against perceived risk.

4.2. Acceptance

Most policies in the sample (78%) were seemingly accepted by mass publics. Importantly, the category of acceptance includes cases without substantial disapproval and without noticeable mobilization (Figure 1) and therefore acceptance could be active or passive (see Section 2.2). Nonetheless, the overall pattern suggests that hard policies might often be adopted without either major support or contention among mass publics. Interestingly, the type of policy instrument does not seem to delimit acceptance because a substantial number of both regulatory and pricing/taxation policies are reflected. This is perhaps surprising given that hard policies are often thought to be contentious, especially those involving pricing/taxation. But it also raises questions about whether these policies are specifically viewed as favourable, people are not aware of the policy being introduced, or because ambition is too low to generate attention.

15 cases were positively categorized as acceptance based on available newspaper data. This included five regulatory policies (Australia, Canada, Colombia, Germany, Mexico), nine pricing/taxation policies (Switzerland, Colombia, Germany, Denmark, France, UK, Mexico), and one policy regarding coal phase-out in Chile. In turn, 24 cases were categorized as acceptance by default due to an absence of evidence of contention in the data. This included eight regulatory policies (Austria, Canada, Japan, Poland, Sweden, the USA), 14 pricing/taxation policies (Chile, Spain, France, the UK, Ireland, Japan, South Korea, Turkey), and two phase-out policies related to exploration and exploitation of fossil fuels in Costa Rica and France.

Some policies categorized as accepted are framework policies. For example, the UK Finance Act 2011 contains over 200 provisions and grants 'certain duties, alters other duties, and to amend the law relating to the National Debt and the Public Revenue, and to make further provision in connection with finance' (Chapter 11, UK Finance Act 2011). Yet, their broad scope might cover up more specific policy measures that are potentially contentious. For example, one of the provisions in this Act was an increase in carbon change levy (§23, Part 1) that largely went unnoticed in Part 6 'Other taxes'. Hence, such policies might impact public responses regarding other more specific policies.

Sometimes contention occurred within a specific industry sector or decision arena (e.g. ministry, sub-national government). However, if no mass public response followed, we treated these as acceptance (i.e. ordinary disagreement eventually resulting in broad public acceptance). For example, in the case of Chile's Carbon Tax policy (2014) protests arose from a particular agricultural sector which led to a series of governmental negotiations with industry. These measures as well as the refined policy structure helped avoid the spillover of contention into the wider public domain (Musquiz, 2018). In the case of France's Excise Tax on Energy Products (2011) contention arose among different levels of authority (i.e. municipal, departmental, national) regarding the beneficiaries of excise revenue but this did not evolve into wider public discontent.

The absence of mobilization concerning policies could be due to an expectation among mass publics that policy adoption has been successfully won. For example, Germany's Fuel Emissions Trading Act (2019) experienced a high degree of consensus and therefore was not an enduring object of contention. On the other hand, Mexico's Emissions Trading Scheme Pilot Project (2020) was deliberately positioned as a pilot to reduce the risk of contention. And in the UK, the public largely remained indifferent to the Finance Act (2011) containing climate policy provisions due to the broad nature of this policy. Hence, acceptance may arise actively or passively.

4.3. Non-acceptance

Two policies were categorized as non-acceptance: the UK Energy Acts of 2011 and 2013. Both cases involved negative public responses due to *unambitious* climate policy. The Energy Act 2011 established a poorly enforced obligation on energy companies to help certain groups of consumers to save energy. However, dissatisfaction arose when payments to households for energy efficiency improvements were insufficient due to purportedly high costs, which resulted in additional costs to households and growing concerns about energy poverty. The Energy Act 2013 introduced an emissions performance standard for new power plants, but this was much lower than understood to be needed to meet climate targets and allowed new gas plants to lock in high levels of emissions into the future. This was criticized by environmentalists and provoked heated parliamentary debate and public frustration. Nonetheless, in both cases specific mobilization against the policies appeared to be missing. Disapproval mostly took form of negative sentiment about climate and energy policy (James, 2023).

4.4. Backlash

Six policies encountered backlash from mass publics, including all three types of policy instruments. Yet, four cases involved carbon pricing/taxation (i.e. Australia, Canada, France, and Mexico). France's Carbon Charge Component was gradually increasing since 2014, yet major public discontent erupted in 2018–2019 in the form of the Yellow Vests Movement when policy-associated costs became perceived by many as excessive and unfair for those already experiencing economic anxiety particularly in non-urban areas. Mexico's Carbon Tax introduced in 2014 led to heated protests later in 2017 when rising fuel prices contributed to energy poverty among nearly 37% of households (Gutiérrez, 2017). In contrast, Australia's Carbon Pricing Mechanism (2011–2012) and Canada's Fuel Charge and Pollution Pricing Act (2017–2019) both faced immediate contention and were exploited by political opposition as a divisive issue despite rebates and compensation mechanisms for low-income households.

The only regulatory policy case of backlash was the US Affordable Clean Energy Rule (2019) which encountered a high level of contention because it lowered environment protection standards from the prior Obama-era Clean Power Plan. Political partisanship played a crucial role in stirring public debate and mobilizing opposition to the rule which led to a series of nationwide protests and helped to put this regulation on hold. The only phase-out policy case of backlash was Germany's Coal Exit Law (2020) which also faced public discontent, but this was due to a perceived insufficiently low level of climate ambition which led to nationwide protests in support of more ambitious climate policy (Haas et al., 2022).

These cases show different reasons for backlash against climate policy. In France and Mexico, costs imposed on people unable to shoulder them provoked intense contention. However, in Australia and Canada, the drivers of backlash appear to be more centrally related to significant societal polarization and divisive populism opposing climate policy. Differently again, backlash arose in Germany in response to policies perceived as too weak rather than too costly. This shows that backlash is not exclusively a reaction against any climate policy but might arise because of a perception of too much or too little ambition.

5. Discussion

To examine the observed patterns of public responses, we consider potential links between public responses and four policy attributes: instrument type, gradual policy introduction, country, and timing. First, instrument

type does not seem to produce consistent public responses. For instance, carbon pricing policies, which are often considered to be unpopular among mass publics (Hammerle et al., 2021; Klenert et al., 2018; Skovgaard & Ferrari, 2023), were not exclusively contested and were often even accepted. This instrument type encountered strong immediate opposition in Canada and Australia, delayed opposition in France, and was accepted in Switzerland⁷, Chile, Colombia (where further tax increase was not applied). However, there were no cases of support for carbon pricing. Regulation was the only policy instrument type that encountered all four public responses. Although, this may simply reflect variation in regulatory policies themselves (Hughes & Urpelainen, 2015; OECD, 2022a; Pacheco-Vega, 2020). Despite being the least represented policy instrument, phase-out policies cover diverse issue areas (from nuclear to coal) with mixed results, which could be because public responses are highly context dependent (Egli et al., 2022; Rinscheid et al., 2022).

Second, policies with a gradual introduction (ramping up or tightening over time) had varying public responses. In some instances (e.g. Swiss CO₂ Compensation Duty (2014)), policy recipients grew accustomed to the regulation and largely accepted progressive increases (Levin et al., 2012; Thalmann & Baranzini, 2006). Yet, in the case of Carbon Charge Component in France (2014), a progressively increasing policy burden, linked to wider concerns and grievances, resulted in major backlash in 2018 (Driscoll, 2023). This cautions against seeing incremental ramping-up of policy as a solution to prevent contention. Policies with redistribution also had varying public responses. In the case of Canada's carbon pricing policy package, a rebate scheme announced in 2018 seemed to reduce disapproval (Toronto Star, 2019) at least in the short-term, although its longer-term effects remain unclear and possibly more shaky (Mildenberger et al., 2022). In the case of Australia's Carbon Pricing Mechanism, redistribution through tax cuts for low-income households and direct payments to low- and middle-income households did not outweigh perceptions of a cost burden, and thus did not allay contention. This suggests that redistribution even if progressive and/or premised on fairness (Carattini et al., 2018; Maestre-Andrés et al., 2021) may not always mitigate contention.

Third, there does not seem to be clear geographical patterns in mass public responses. Although, in some countries, diverging public responses such as backlash and support (e.g. Mexico, USA) or acceptance and non-acceptance (e.g. UK) were present across different policies. This might suggest a role for political system type in influencing public responses as discussed in the literature. For example, scholars have argued that majoritarian systems can foster division between political parties and publics over climate policy due to incentives for political opportunity (Andersen, 2019; Lipsy, 2018; Mildenberger, 2020).

Fourth, temporally, there appears to be an increase in contention in the post-2015 period. This might be simply due to an increase in the number of adopted policies, or perhaps an increase in stringency of proposed climate policies leading to increased public contention (Botta & Koźluk, 2014; Glaus, 2021; Nachtigall et al., 2022). However, to assess this, in-depth analysis of stringency would be needed (e.g. Schaffrin et al., 2015). It is also important to examine the pre-adoption stage of policy which is likely to reveal substantially more contention over policy formulation and adoption. Furthermore, timing might play a role in individual policy cases. For instance, timing of policy adoption vis-à-vis electoral cycles or broader socioeconomic developments (e.g. economic crises, societal polarization) could matter (Brown & Sovacool, 2017; Farstad et al., 2018; Hutter & Weisskircher, 2023).

Overall, the finding of a relatively low prevalence of contentious public responses in the sample (total of 16%, comprising 12% backlash and 4% non-acceptance) is noteworthy given the general expectation among scholars and policymakers that hard policies are prone to contention and backlash. One explanation for the seemingly low prevalence of contentious public responses could be a relatively low level of ambition within the sampled policies. In other words, policy recipients are not excessively burdened by such policies. Another explanation could be unawareness by mass publics about the policies. This could be intentional by policymakers (e.g. policy discussion is minimized or introduced at times where other issues are more salient), structural (e.g. policy is created through inter-party negotiation in ways that make attribution of responsibility unclear) or incidental (e.g. due to genuinely low level of climate-related concerns). On the other hand, a more positive interpretation could be that latent public desire for climate policy is higher than typically thought which moderates contentious public responses. But this is less likely since we would then also expect to see greater contention over lack of ambition, because our approach focuses on detecting contention regardless of direction (i.e. too high or too low ambition). Reasons for the occurrence of contentious

public responses may also vary across different cases due to the embeddedness of specific policies within complex socio-political contexts where they become objects of attention, including in relation to other issues and debates (e.g. sensitivity to costs, conservative or progressive shifts in public mood). This would suggest that a key issue is how climate policy calibrates to specific socio-political contexts and moments in time.

6. Conclusion

We contribute to work on ex-post analysis of public responses, complementary to ex-ante studies of public support, both of which are important given the challenges of advancing ambitious and durable climate policy and the potential causal complexity of public responses. Policy struggles do not end with policy adoption, and our study suggests that policymakers must remain attentive to post-adoption contention and the role of mass publics at this stage. Although our findings reveal a relatively low prevalence of contention at the post-adoption policy stage, the chances of such contention may grow in the future both as climate policy becomes more demanding and due to wider fractious contemporary politics. Nonetheless, politicians and policymakers should not be axiomatically fearful about the lack of public acceptance of hard climate policies. There may be more latitude than commonly thought to advance hard climate policies, especially as climate policy debates keep rapidly evolving and the realities of unfolding energy transitions become increasingly apparent within society and politics.

Several policy implications can be drawn. First, policy makers should consider stringency not only as a matter of policy design (*de jure*) but also as it manifests contextually (*de facto*). For example, the incremental increase of France's carbon charge component (initially set out in 2014 but triggering a contentious public response several years later) coincided with growing economic insecurity for many people, meaning that policy stringency was perceived differently in peoples' lives than it may have seemed in design alone. Relatedly, hard climate policy could include flexibility to moderate policy stringency in changing contexts and to soften requirements when unexpected pressures arise. Policy moderation is already sometimes done in ad-hoc ways (such as when politicians suspend charges or obligations in the face of criticism), but this could be structurally designed-in to enable credible proactive responses following pre-specified rules. It could also include the opportunity to increase policy stringency when circumstances allow, such as when renewable energy targets are met sooner than anticipated, when a boost is needed to achieve pre-defined targets, or when economic or social circumstances allow the possibility to ramp up ambition. This resonates with broader calls by some policy scholars for attention to ways of 'moderating the policy thermostat' (Cashore & Howlett, 2007; Rosenbloom et al., 2019). However, such an approach could also risk tampering by politicians wishing to hold back climate policy, emphasising the need for credible moderation procedures. Lastly, policy makers might consider trying to keep policy salience low. However, the efficacy of such an approach is not clear since policy developments are likely to become apparent anyway risking greater criticism, and ethical questions also arise over transparency. Thus addressing public concerns thoroughly within policy design is likely to be a more robust approach.

Crucially, our study does not cover cases of policy proposals which experience contention (and might be rejected) *prior to* adoption. This is because our focus was on the post-adoption stage as a first step that can be expanded in future work. Our goal was to assess prevalence using an objective sample frame to avoid sampling on the dependent variable. As such, the analysis provides a baseline assessment for the post-adoption phase only. We also did not study sub-national levels of climate policy, yet this would be interesting and important to examine and bring alongside national-level analysis especially because it is often thought that states/provinces/regions can exert counter-veiling policy directionality compared to the national level.

An important next step for future research is to analyze the drivers of contentious public responses to hard climate policies. This could include factors relating to policy characteristics but also institutional and societal context. For example, design features such as redistribution and progressive phase-in seem to have mixed results concerning public responses, and other design factors (e.g. timing, sequencing) need further work and possibly also larger samples to scrutinize. Yet, Mildenberger et al. (2022) find that people may respond in non-rational ways (i.e. unaligned with actual policy design) due to partisan cues that can bias peoples'

responses to policy. Others point to the political contexts under which policy is adopted (e.g. interest intermediation, electoral blame shielding) as important in understanding contention (Andersen, 2019; Finnegan, 2022). Further institutional factors (e.g. election proximity) and non-institutional factors (e.g. prior mobilization, economic crisis, cultural grievance) might also influence public responses. This points to a complex task to explain public responses, which could arise through different causal mechanisms. Our study provides a first step towards such work by identifying and reflecting on patterns of public responses to inform future analysis of factors and mechanisms causing them.

Notes

1. Although, we also acknowledge that policies often come as a mix (Glaus, 2021; Howlett & Rayner, 2007).
2. For example, backlash could occur due to policy perceived as too strong or too weak, an issue we may see grow in coming years.
3. This is especially important for general frameworks which might contain multiple instrument types. Therefore, in data collection and analysis, we screened newspaper articles for public responses specifically linked to hard climate policy elements.
4. By “newspaper of quality” (also referred in the literature as “newspaper of record”), we mean a newspaper with “a certain reputation for consistent attention to accuracy and depth in reporting news” (Martin & Hansen, 1996, p. 591).
5. For example, well known databases as Mass Mobilization Database, Global Protest Tracker, Nonviolent and Violent Campaigns and Outcomes, and ACLED (Armed Conflict Location & Event Data) use newspaper data as sole or key sources.
6. Formally referred to as “Abrogation of the legal regime applying to nuclear energy production” (2011).
7. This refers to the so-called “second CO₂ Tax” (2013) which included carbon trading mechanisms. This policy was accepted by the population (or at least, no evidence of contention was detected as per the coding procedure). Later unsuccessful attempts to revise the law (e.g. 2021 referendum) are not part of the sample due to our focus on adopted policies. Yet, it is important to state that a carbon tax remains contested in the Swiss context.

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